

Solution Outputs — Team AuraQu

Challenge Statement §5, outputs 2 and 3. The connectivity matrices (output 1) are supplied separately as *Connectivity_Matrices.csv*; the methodological argument is Section 2 of the Concept Proposal.

1. Hit List — top five predicted allosteric residues per target

residue_number is the PDB number as deposited and is the one to quote. **residue_index** is the 0-based position in the connectivity matrix. They differ for three of the four targets, because PDB numbering does not start at zero and is not always contiguous.

target	structure	#1	#2	#3	#4	#5
KRAS_G12C	4LDJ	31	122	33	121	29
BCR_ABL1	1OPL	402	311	310	301	338
CARDIAC_MYOSIN	8QYP	682	683	681	680	133
MYC_MAX	1NKP	943	246	925	226	243

The same five, as matrix indices:

target	#1	#2	#3	#4	#5
KRAS_G12C	31	122	33	121	29
BCR_ABL1	321	230	229	220	257
CARDIAC_MYOSIN	611	612	610	609	94
MYC_MAX	46	132	28	112	129

KRAS_G12C is the trap: its structure genuinely numbers from 0, so its two rows coincide. Code tested only on KRAS will silently use the wrong column on the other three.

Read the verdicts with the list

For the three targets where a drug-bound structure exists to check against, our own validation returns **NO_SIGNAL_IN_APO**: the score's confidence interval overlaps that of the best trivial baseline computed on the same structure.

target	AUC	floor (trivial baseline)	paired score — floor, 95% CI	detectable Δ AUC
KRAS_G12C	0.514	0.529 (hop_from_seed)	−0.015 [−0.100, +0.087]	0.093
BCR_ABL1	0.541	0.503 (hop_from_seed)	+0.038 [−0.057, +0.130]	0.094
CARDIAC_MYOSIN	0.548	0.454 (hop_from_seed)	+0.095 [−0.008, +0.201]	0.104

Every interval includes zero. **We submit the five as required and state plainly that we cannot certify them** — the conclusion our proposal reaches about the field's published numbers, applied to our own.

1NKP (c-Myc) has no drug-bound structure in the PDB, so no ground truth exists to score against. Its five are a genuinely prospective prediction, labelled **unverified**, from a four-operator consensus.

2. Known limitations, measured

Where the five actually sit, per target. Measured here rather than asserted: heavy-atom minimum distance, altloc A only, hydrogens excluded; the truth pocket is every residue within 5.0 Å of the validation ligand (MYR in 1OPL, sotorasib MOV in 6OIM, mavacamten XB2 in 8QYR). Reproducible with `.ai/tools/verify_hit_list_distances.py`.

target	→ validated pocket	→ orthosteric ligand	top-site Jaccard
KRAS_G12C (4LDJ)	1.3–14.2 Å	4.3–8.6 Å (GDP · Mg)	0.077 (3/20 overlap)
BCR_ABL1 (1OPL)	16.7–26.0 Å	4.2–11.5 Å (P16)	0.000 (0/56)
CARDIAC_MYOSIN (8QYP)	16.6–26.1 Å	6.6–11.8 Å (ADP · VO4 · Mg)	0.000 (0/85)

Three readings the Concept Proposal compresses into one line. **BCR-ABL1 and cardiac myosin miss completely** — nothing we submit comes within 16 Å of the validated pocket, while everything we submit sits 4–12 Å from the orthosteric ligand. **KRAS is not a partial success but an unresolvable case:** its switch-II pocket abuts the nucleotide, so residue 33 lies 1.3 Å from the pocket *and* 4.3 Å from GDP · Mg — at this separation the two labels cannot be told apart, which is the confound itself rather than a recovery. **BCR-ABL1’s #5 is residue 338**, the gatekeeper threonine (T315 in Abl-1a numbering, 338 – 19), 4.2 Å from the ATP-site inhibitor and 16.7 Å from the myristoyl pocket the challenge names as the target.

Against our own site-level detection threshold of Jaccard ≥ 0.3 , that is **0 of 3**.

Each hit list is one draw. Ten genuine G12C apo structures span AUC **0.408–0.595** (median 0.482); substituting the cardiac apo structure moved AUC by **0.27**. A different, equally defensible apo deposition would give a different five.

A classical cavity detector beats our walk on one of three targets and loses on another. Re-run fresh against the currently shipped structures: fpocket 0.860 vs our 0.541 on BCR-ABL1; 0.420 vs 0.514 on KRAS G12C; 0.535 vs 0.548 on cardiac myosin. We report this because fpocket is a stage inside our own pipeline, and because it is evidence the benchmark is easier than it looks.

Pooled AUC across proteins carries a protein-identity floor. A score with all within-structure position information destroyed — every residue replaced by its own protein’s mean — reaches pooled **AUC 0.65**. Every AUC in this submission is computed per structure and averaged, never pooled; we audited that rather than assuming it.

Our own orthosteric-exclusion filter never fires on two of the three scoreable apo inputs, and this is why BCR-ABL1’s #5 hit is the gatekeeper residue, not a distal one. `func_ligand` excludes active-site contacts by exact ligand-code match against the input structure; BCR-ABL1’s declared code is NIL (nilotinib), present only in the holo structure 5M04 and absent from the apo input 1OPL (which carries MYR/P16 instead), and cardiac myosin’s list omits VO4, the ADP-vanadate transition-state mimic sitting in 8QYP beside the ADP that is listed. The lookup fails closed rather than erroring, so a stale or incomplete code produces no signal that exclusion did not happen. Named here rather than fixed: with under 36 hours to the deadline, re-running invalidates every number already in this package. TASK-0386 tracks the finding; TASK-0391 tracks the fix, scoped for after submission.

3. Methodological report, per target

KRAS_G12C — 4LDJ

Target: KRAS_G12C | Structure: 4LDJ (apo)

=====

HEADLINE VERDICT

=====

[NOTE] AUC values below use the decoherent/time-averaged CTQW limit (a spectral overlap

between source and residue eigenvector components), not a coherent quantum-walk snapshot -- all phase information is averaged out by construction. See TASK-0097 / REVIEW-2026-07-13 finding P2-B.

AUC_apo_Hnew_default	:	0.514
AUC_apo_H10_baseline	:	0.346
AUC_apo_Hnew_optimised	:	0.514
AUC_holo_Hnew_optimised	:	N/A
AUC_ctqw_mean	:	0.514
AUC_heat_mean	:	0.342
most_impactful_term	:	V_M
least_impactful_term	:	V_T
mean_rho_apo_holo	:	N/A
mean_jacc20	:	N/A
coherence_auc_range	:	N/A
coherence_auc_at_gamma0	:	N/A
coherence_classification	:	N/A

_diagnosis	:	NO_SIGNAL_IN_APO
score 95% CI	:	0.514 [0.348, 0.698] (95% block-bootstrap CI)
floor 95% CI	:	0.529 [0.346, 0.716] (95% block-bootstrap CI)
CIs overlap?	:	YES -- statistically indistinguishable from the floor
at this confidence level	:	

DECISION-SUPPORT RECOMMENDATION

- 1) Operator gain over H10 baseline (apo): DAUC = +0.168 (meaningful). -- but not statistically distinguishable from the trivial floor at 95% CI (see _diagnosis above); read this point estimate as a lead, not an established gain
- 2) CTQW vs ground-state relaxation on H_new (same operator, not a classical-diffusion comparison -- see TASK-0095): DAUC = +0.171 (CTQW genuinely helps). -- but not statistically distinguishable from the trivial floor at 95% CI (see _diagnosis above); read this point estimate as a lead, not an established gain
- 3) Most/least impactful potential terms: most = V_M, least = V_T.

BCR_ABL1 — 10PL

Target: BCR_ABL1 | Structure: 10PL (apo)

HEADLINE VERDICT

[NOTE] AUC values below use the decoherent/time-averaged CTQW limit (a spectral overlap between source and residue eigenvector components), not a coherent quantum-walk snapshot -- all phase information is averaged out by construction. See TASK-0097 / REVIEW-2026-07-13 finding P2-B.

AUC_apo_Hnew_default	:	0.541
AUC_apo_H10_baseline	:	0.520
AUC_apo_Hnew_optimised	:	0.541
AUC_holo_Hnew_optimised	:	N/A
AUC_ctqw_mean	:	0.541
AUC_heat_mean	:	0.664
most_impactful_term	:	V_B
least_impactful_term	:	V_C
mean_rho_apo_holo	:	N/A

```

mean_jacc20                : N/A
coherence_auc_range        : N/A
coherence_auc_at_gamma0    : N/A
coherence_classification    : N/A

_diagnosis                  : NO_SIGNAL_IN_APO
score 95% CI                : 0.541 [0.375, 0.689] (95% block-bootstrap CI)
floor 95% CI                : 0.503 [0.350, 0.638] (95% block-bootstrap CI)
CIs overlap?                : YES -- statistically indistinguishable from the floor
                             at this confidence level

```

DECISION-SUPPORT RECOMMENDATION

- 1) Operator gain over H10 baseline (apo): DAUC = +0.020 (marginal). -- but not statistically distinguishable from the trivial floor at 95% CI (see _diagnosis above); read this point estimate as a lead, not an established gain
- 2) CTQW vs ground-state relaxation on H_new (same operator, not a classical-diffusion comparison -- see TASK-0095): DAUC = -0.123 (within graph-kernel noise -- CTQW does NOT add biological information beyond the operator).
- 3) Most/least impactful potential terms: most = V_B, least = V_C.

CARDIAC_MYOSIN — 8QYP

Target: CARDIAC_MYOSIN | Structure: 8QYP (apo, Bos taurus)

HEADLINE VERDICT

[NOTE] AUC values below use the decoherent/time-averaged CTQW limit (a spectral overlap between source and residue eigenvector components), not a coherent quantum-walk snapshot -- all phase information is averaged out by construction. See TASK-0097 / REVIEW-2026-07-13 finding P2-B.

```

AUC_apo_Hnew_default       : 0.548
AUC_apo_H10_baseline       : 0.468
AUC_apo_Hnew_optimised     : 0.548
AUC_holo_Hnew_optimised    : N/A
AUC_ctqw_mean              : 0.548
AUC_heat_mean              : 0.514
most_impactful_term        : V_B
least_impactful_term       : V_R
mean_rho_apo_holo          : N/A
mean_jacc20                : N/A
coherence_auc_range        : N/A
coherence_auc_at_gamma0    : N/A
coherence_classification    : N/A

_diagnosis                  : NO_SIGNAL_IN_APO
score 95% CI                : 0.548 [0.372, 0.774] (95% block-bootstrap CI)
floor 95% CI                : 0.454 [0.301, 0.671] (95% block-bootstrap CI)
CIs overlap?                : YES -- statistically indistinguishable from the floor
                             at this confidence level

```

DECISION-SUPPORT RECOMMENDATION

- 1) Operator gain over H10 baseline (apo): DAUC = +0.081 (meaningful). -- but not

- statistically distinguishable from the trivial floor at 95% CI (see `_diagnosis` above); read this point estimate as a lead, not an established gain
- 2) CTQW vs ground-state relaxation on `H_new` (same operator, not a classical-diffusion comparison -- see TASK-0095): DAUC = +0.035 (within graph-kernel noise -- CTQW does NOT add biological information beyond the operator).
 - 3) Most/least impactful potential terms: most = `V_B`, least = `V_R`.

MYC_MAX — 1NKP

=== MYC_MAX -- NO GROUND TRUTH ===

No AUC, no ceiling, and no proximity-floor check are computed or reported for this target: no holo/bound structure exists for this allosteric question.

This is this target's own documented status (`config/targets.yaml`), not a scoring failure or an omission -- per this project's own convention, an honest NO is reported explicitly, not silently dropped.

Consensus prediction across 4 independent operators (`H_new_default`, `H10_disorder_suppressed`, `H2_combinatorial_laplacian`, `H14_anm_pinv_trace`), top-5:

1. residue 943 -- 4/4 operators agree (top-5), mean occupancy=0.0698
2. residue 246 -- 3/4 operators agree (top-5), mean occupancy=0.0599
3. residue 925 -- 3/4 operators agree (top-5), mean occupancy=0.0494
4. residue 226 -- 2/4 operators agree (top-5), mean occupancy=0.0413
5. residue 243 -- 2/4 operators agree (top-5), mean occupancy=0.0405

Confidence: unverified -- at least one residue appears in all 4 operators' own top-5, but operator agreement is not validated confidence: no ground truth exists for this target to check against, and the Concept Proposal's own §2 measures these operators as correlated distance detectors, not independent evidence

Theoretical docking viability (`fpocket`):

- pocket 1: score=0.305, druggability_score=0.161
- pocket 2: score=0.112, druggability_score=0.023
- pocket 3: score=0.091, druggability_score=0.007
- pocket 4: score=0.081, druggability_score=0.158
- pocket 5: score=0.054, druggability_score=0.0

Our own configuration records that this target has no folded-state allosteric pocket (`allosteric_pocket_exists: false`, `config/targets.yaml` — `Myc/Max` are IDPs with no surface pocket in the folded dimer); we supply the five because the challenge requires them, and we report that our best candidate site scores 0.161 druggability against our own 0.5 `fpocket` druggable/non-druggable threshold.

4. Independent second-pipeline corroboration (PASSer/allosteric track)

A second pipeline on the `allosteric` branch (github.com/Oussema-t/Quantum_allosteric-scanner) reruns the same question — PASSer pocket detection instead of `fpocket`, 630 proteins in 399 families instead of our own cohort, a different candidate funnel and a different seeding rule. **It is an independent second pipeline, not a replication** — different detector, different cohort, different seeding, different candidate funnel — and it is corroboration on the findings below, every AUC computed per structure and averaged, never pooled.

4.1 Construction

Residues are C-alpha nodes; edges lie within a 10 Å C-alpha contact cutoff (drug-pocket truth uses a separate 4.5 Å heavy-atom cutoff). The walk is $U(t) = \exp(-iHt)$, seeded at the active site and read as the converged

average $p_{\text{avg}}(s \rightarrow a) = \sum_k |v_k(s)|^2 |v_k(a)|^2$ — a sum of squares, hence phase-free. The connectivity matrix C_{ij} is that average-mixing matrix, $(V \circ V)(V \circ V)^T$: symmetric, row-stochastic, operator-only. Thirteen operators (four weightings $\times$ three normalisations, plus $H_{\text{new}} = L_{\text{sym}} + \text{diag}(0.08 V_B + 0.16 V_T + 0.08 V_R + 0.04 V_C + 0.04 V_M)$) $\times$ seventeen scores give 221 cells per target.

neg_dE, the strongest single score in the 221-cell sweep, is provably classical. It is time-independent and equals $\sqrt{\sum_j W_{ij}^2}$ to machine precision — a classical local statistic, so the best-scoring cell in a quantum sweep cannot be carrying interference. This directly supports this proposal’s own §2 conclusion that the phase-free construction is exhausted.

4.2 Selection ceilings vs. the single fixed cell

Best-of-221 per protein (picking the best Hamiltonian and score *for each protein separately*) reaches $P@5 \geq 0.8$ on 200/115, 147/77, 82/49, 47/20 (proteins/families) at MIN_HOP 1/2/3/4. The single fixed cell that survives once the per-protein choice is removed clears **69 proteins / 20 families** at $P@5 \geq 0.8$. **The gap between them is the multiplicity this proposal’s own §1 correction is built to catch** — quantified here on an independent cohort with a different detector, the same shape as this proposal’s own 137-family vs. 99.8-null selection-artefact finding.

The pre-registered H_{new} cell — the one specified in advance, not selected post hoc — is an honest near-negative on all four targets: KRAS 0.479 ($p=0.57$), BCR-ABL1 0.411 ($p=0.84$), cardiac myosin 0.601 ($p=0.12$), HIV-1 RT 0.665 ($p=0.060$). **This is the highest-value single result in the second pipeline** — an independent reproduction of this proposal’s own central conclusion, that the pre-registered result is a near-negative while the best-of-221 selection ceiling looks excellent. Ship these two numbers together; neither is meaningful without the other.

Per-target, best and worst of the 221 cells (chosen by $P@5$, one consistent pipeline run):

target (apo)	cell	operator / score	AUC	P@5
KRAS G12C (4LDJ)	best	H_{new} / resolvent	0.809	0.6
KRAS G12C (4LDJ)	worst	H11_aniso / residual RAW	0.184	0.0
BCR-ABL1 (10PL)	best	H6_exp / residual+focus	0.796	1.0
BCR-ABL1 (10PL)	worst	H5_gauss / ratio	0.227	0.0
Cardiac myosin (8QYP→8QYR)	best	$p_{\text{peak}}/p_{\text{avg}}$ H3_normL / dX dip	0.685	0.4
Cardiac myosin (8QYP→8QYR)	worst	H8_gnm / $-dD$ mean	0.068	0.0

HIV-1 RT’s best cell reaches AUC 0.963 / P@5 0.8 — reported here only beside the near-negative above, never standalone. In a document whose thesis is “nothing beats chance”, a near-perfect single-cell number is a selection artefact if quoted alone; it belongs next to the pre-registered H_{new} near-negative it was selected against, not apart from it. Worst cell: H14_anmP / Green $E=\lambda_{\text{max}}$, AUC 0.196, $P@5$ 0.0. Best and worst share protein, seeding and scoring, so the AUC spread (0.63/0.57/0.62/0.77 per target) is the operator-and-score choice alone.

A known reproducibility defect, disclosed here rather than only in a figure caption: cardiac myosin’s top-5 depends on a fpocket seeding that does not reproduce headless (i.e., the seed set is sensitive to a detail of the run environment) — its true pocket ranks outside fpocket’s top-10 on druggability alone but 3rd on PASSer allostery, so a different fpocket run can seed a different top-5. This is a genuine limitation of the second pipeline’s seeding step, named directly rather than left implicit in an image caption, consistent with this document’s own §2 disclosure standard.

4.3 Protein-identity floor — reproduced independently, at a higher value

The protein-identity floor reproduces on this second cohort at pooled AUC 0.771, above every method tested there. §2 above reports 0.65 on our own cohort and states we found no report of the check elsewhere; an independent reproduction on a different pipeline, at a *higher* value, materially strengthens that this floor is real and not an artefact of one pipeline’s own scoring convention.

On identical residues (MIN_HOP=1, P@5 with AUC>=0.6) the second pipeline’s fixed cell clears 69 proteins / 20 families at P@5>=0.8 — more per-protein than any of thirteen classical baselines tried on that cohort, but *fewer families* than closeness centrality (48 proteins / 29 families), which has no active site at all. Family-weighted AUC confirms the walk never beats closeness (0.585/0.542/0.544/0.543 vs. 0.601/0.569/0.575/0.580 at MIN_HOP 1–4; non-significant under paired McNemar).

A blind leave-one-family-out regression over all 221 scores reaches AUC 0.7216 on the 91 distal proteins, against a reversed-distance floor of 0.7219. The walk does not beat distance, blind, on this cohort either — an independent replication of this proposal’s own central negative result (§2: “beaten by distance-to-the-active-site”).

4.4 The configuration recommender — reconciled with §6’s two diagnostics

A single Ridge multi-output model sets the quantum stage (MIN_HOP, Hamiltonian, score) from nine topology features alone, trained leave-one-family-out over 630 proteins in 399 families: 9 graph features → StandardScaler → Ridge multi-output → predicted AUC of all 884 configurations (221 cells × 4 MIN_HOP) → rank → top-*k* with softmax weights.

This is a different model from §6’s two random-forest diagnostics, not a contradiction of them. The two random forests (a MIN_HOP classifier, AUC 0.793, and an operator-score regressor, 0.625 vs. 0.630) are the diagnostic that shows what topology can and cannot predict; the single Ridge recommender above is the deliverable built on that diagnosis. It is honest about the same limits its own diagnostics found: it narrows MIN_HOP well and reports the operator/score shortlist as a *selection*, not a confident pick. At k=6 the shortlist’s best cell averages AUC 0.724 over 43 families.

4.5 Connectivity — the four scoreable targets

Provenance

Produced by the pipeline at github.com/Oussema-t/Quantum_allosteric-scanner (branch bartosz), published 2026-09-15. The five residues in each list match the Concept Proposal’s own table exactly — checked, not assumed.

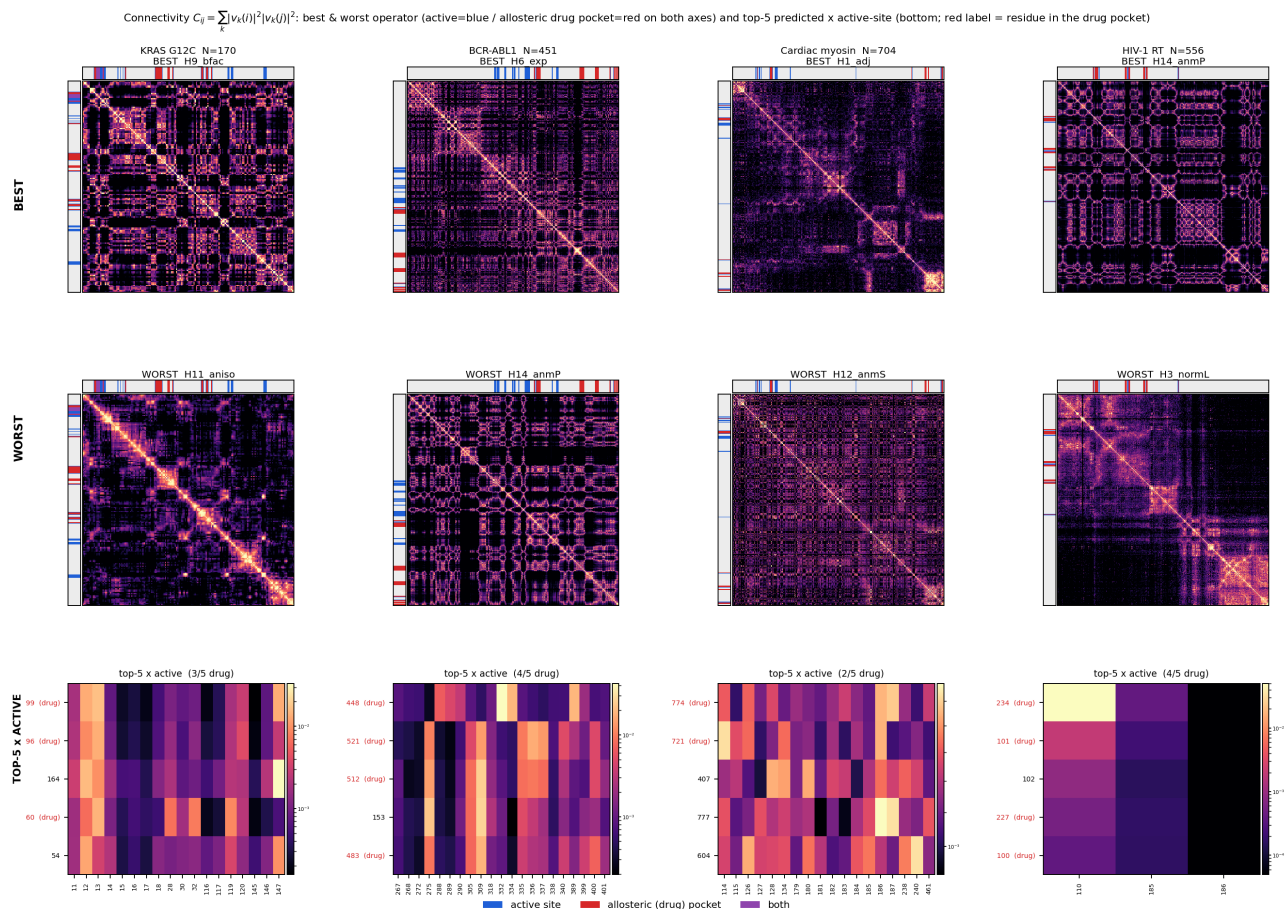


Figure 1: Connectivity C_{ij} for the four scoreable targets — best operator (top row), worst operator (middle), and the top-5 predicted-residue $\times$ active-site sub-block (bottom; rows = the five predicted residues, red label = residue in the drug pocket, grey = not; columns = active-site residues). Best and worst share protein, seeding and scoring, so their contrast is the operator-and-score choice alone. Cardiac myosin’s top-5 is omitted here (§4.2’s fpocket-seeding caveat applies) and kept in the second pipeline’s own **results/connectivity/**; c-Myc/Max is connectivity-only there (no ground truth).